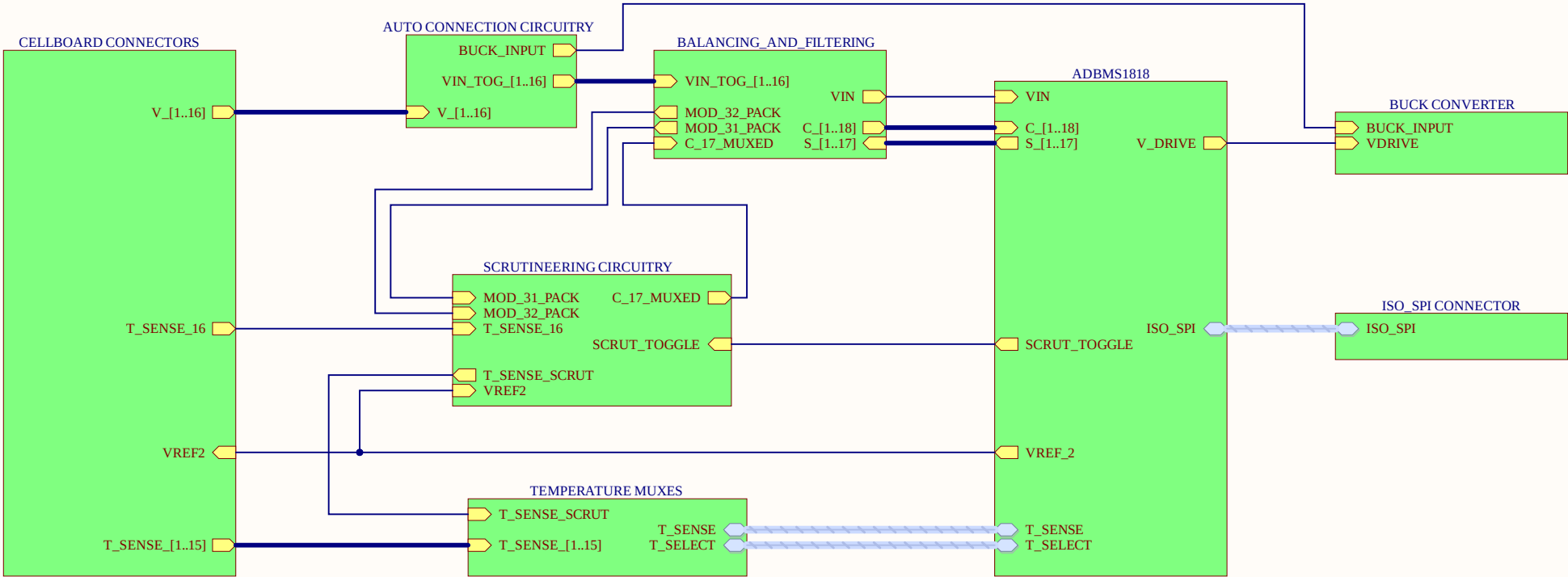


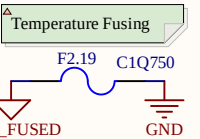
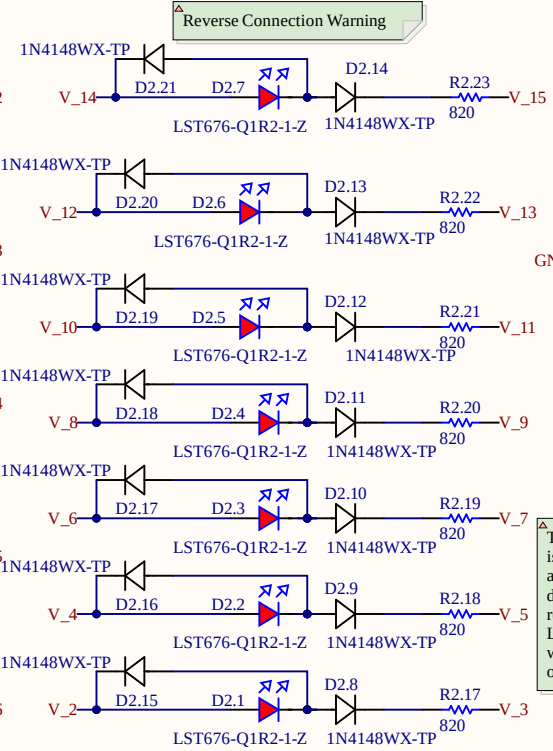
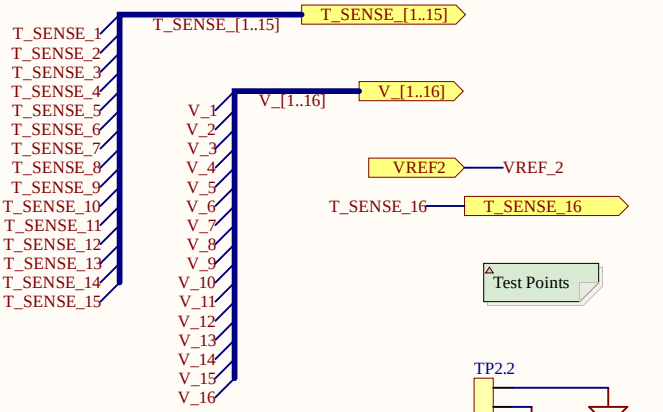
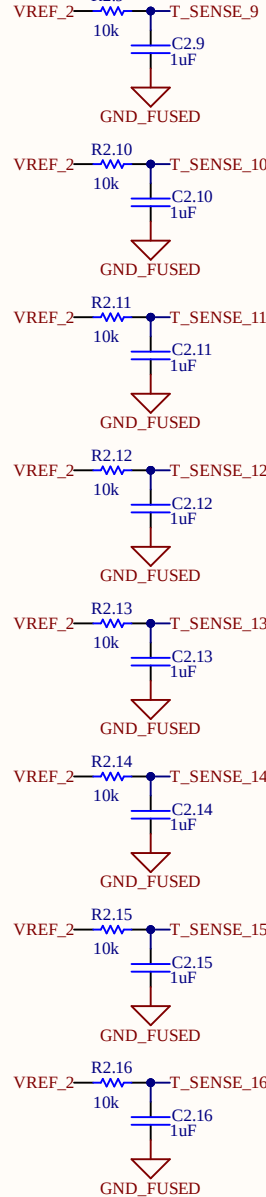
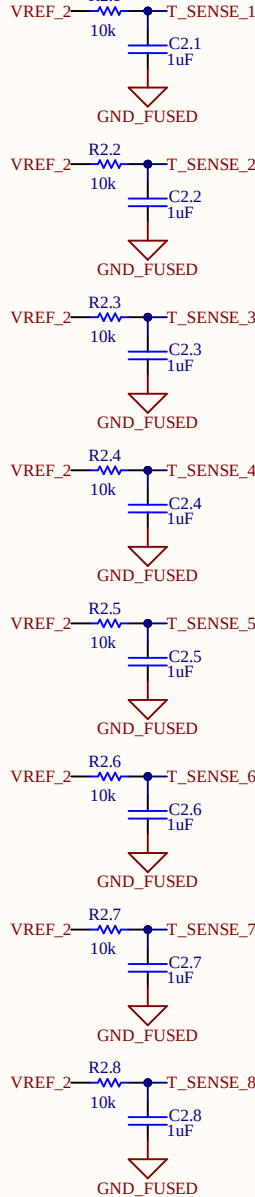
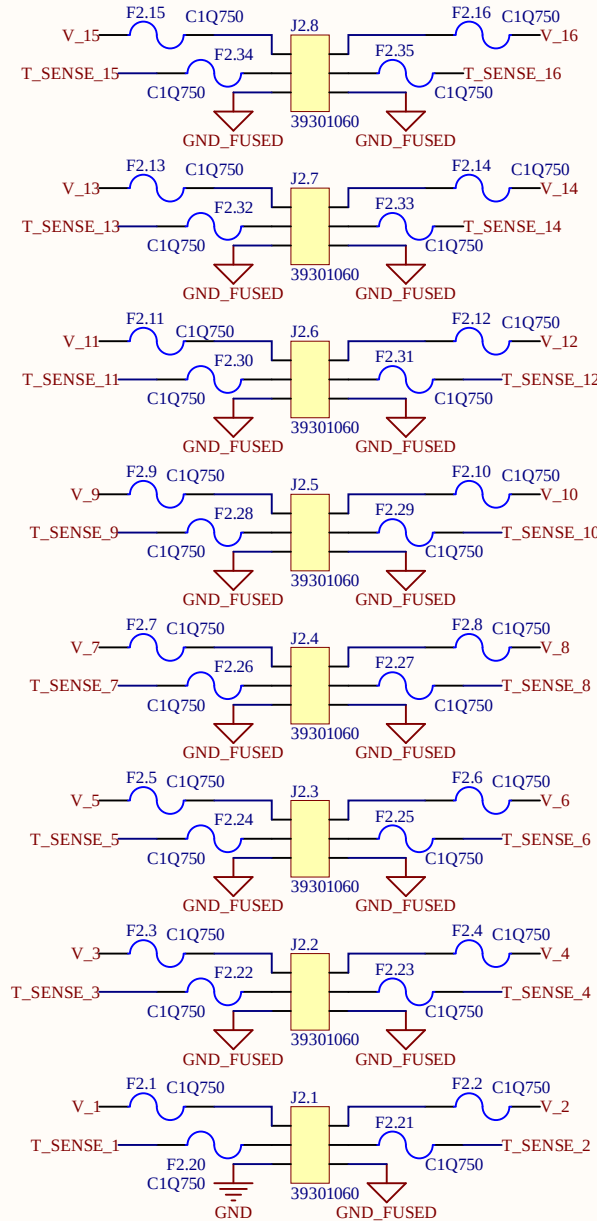
DR DOCUMENT LINK



2. CELLBOARD CONNECTIONS

Module Board Connectors

Temperature Resistor Divider
+ Low Pass Filter



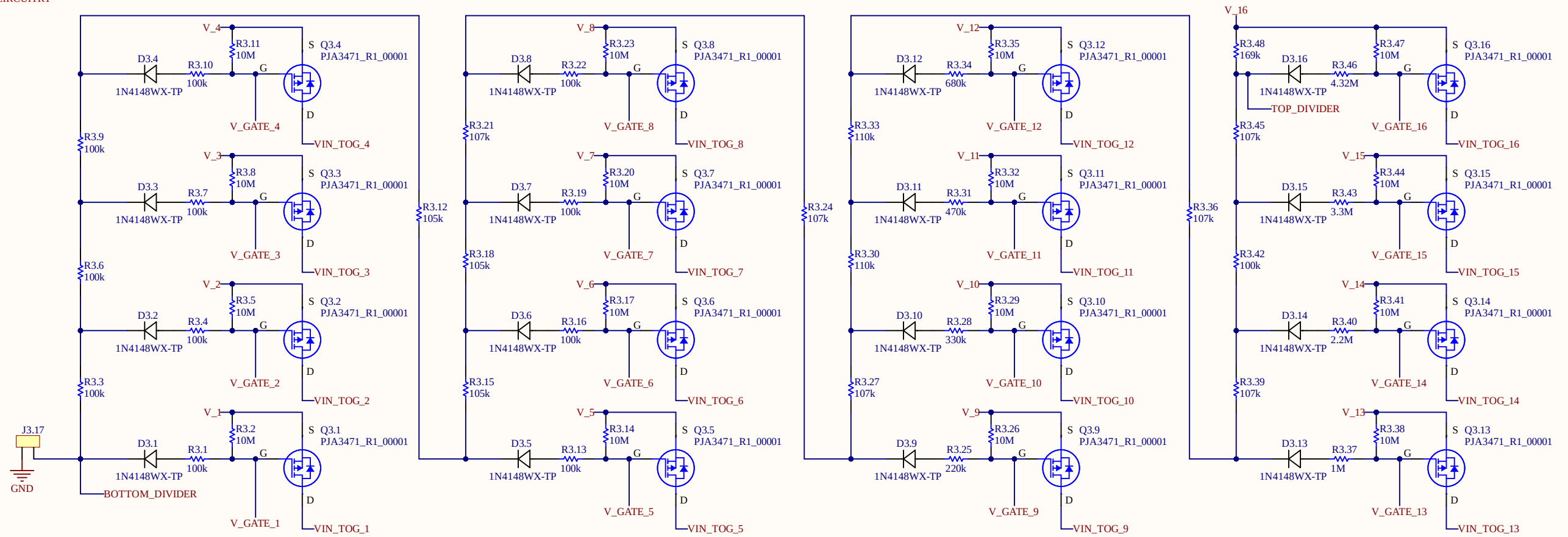
The purpose of the two diodes is to protect the LED against any reverse voltage. The diode in parallel clamps the reverse voltage across the LED, while the diode in series with the LED takes the bulk of the voltage drop.

Each 6-pos connector will attach to 2 different modules (wires will split to 2 separate 4-pos connectors)

These resistors and capacitors were previously on the moduleboards. It serves as a resistor divider when used in series with the thermistor

Title		
Size	Number	Revision
A4		
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File:	Cellboard_Connections.SchDoc	Drawn By:

3. AUTO CONNECTION CIRCUITRY



When the VIN_EN jumper is shorted, the PFETs will close and conduct to the ADBMS. Otherwise, the gates will be pulled up, and the PFETs will open.

When the PFETs are opened (VIN_EN is not grounded), the max voltage across the diodes is ~70V.

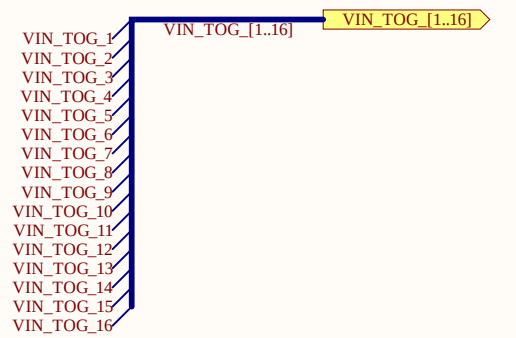
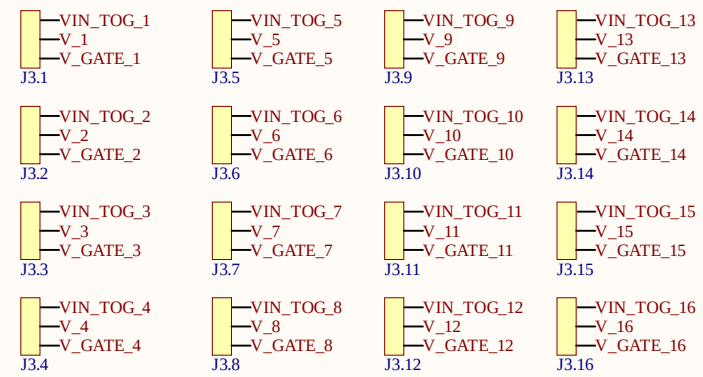
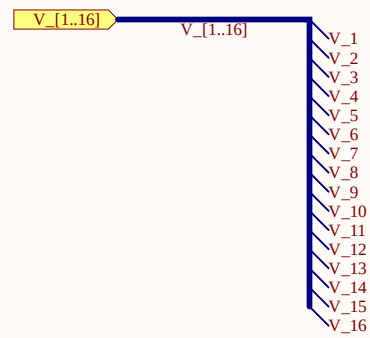
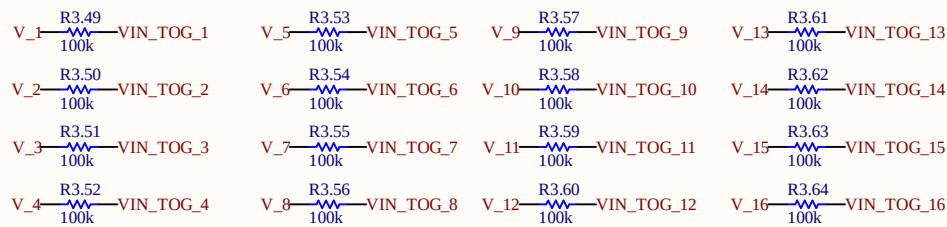
△ The 100k resistors form a large resistor divider from VIN (~70V) down to GND, which allows each PFET gate to be pulled down to a lower voltage using one toggle switch.

10Mega ohm resistors are used to reduce loading the circuit at each point of the resistor divider. If more current was drawn at each node on the resistor divider, then the voltages would be much less evenly split (currently they should have approx the cell voltage across each resistor).

When inserting the module connectors @J3.17 must be left disconnected. Once all module connectors have been inserted, connecting a jumper cap across @J3.17 will cause all the PFETs to begin conducting.

These resistors in parallel with the PFETs, prevent an excessive drain-source voltage buildup on the PFETs allowing us to use PFETs with a smaller max drain-source voltage threshold, which I found to have a smaller on-resistance.

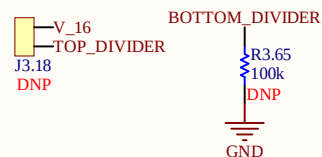
PFET BY-pass resistors
(drain to source)



Auto-Connection Precharge Option (DNP!)

⚠ These components are only to be used if we are using the auto-connection circuitry in "pre-charging" mode.

See DR doc for more info.



ONLY populate two test points (VIN_TOG_X and V_X) for each module section. This will improve the ease of reverting to adding modules by hand.

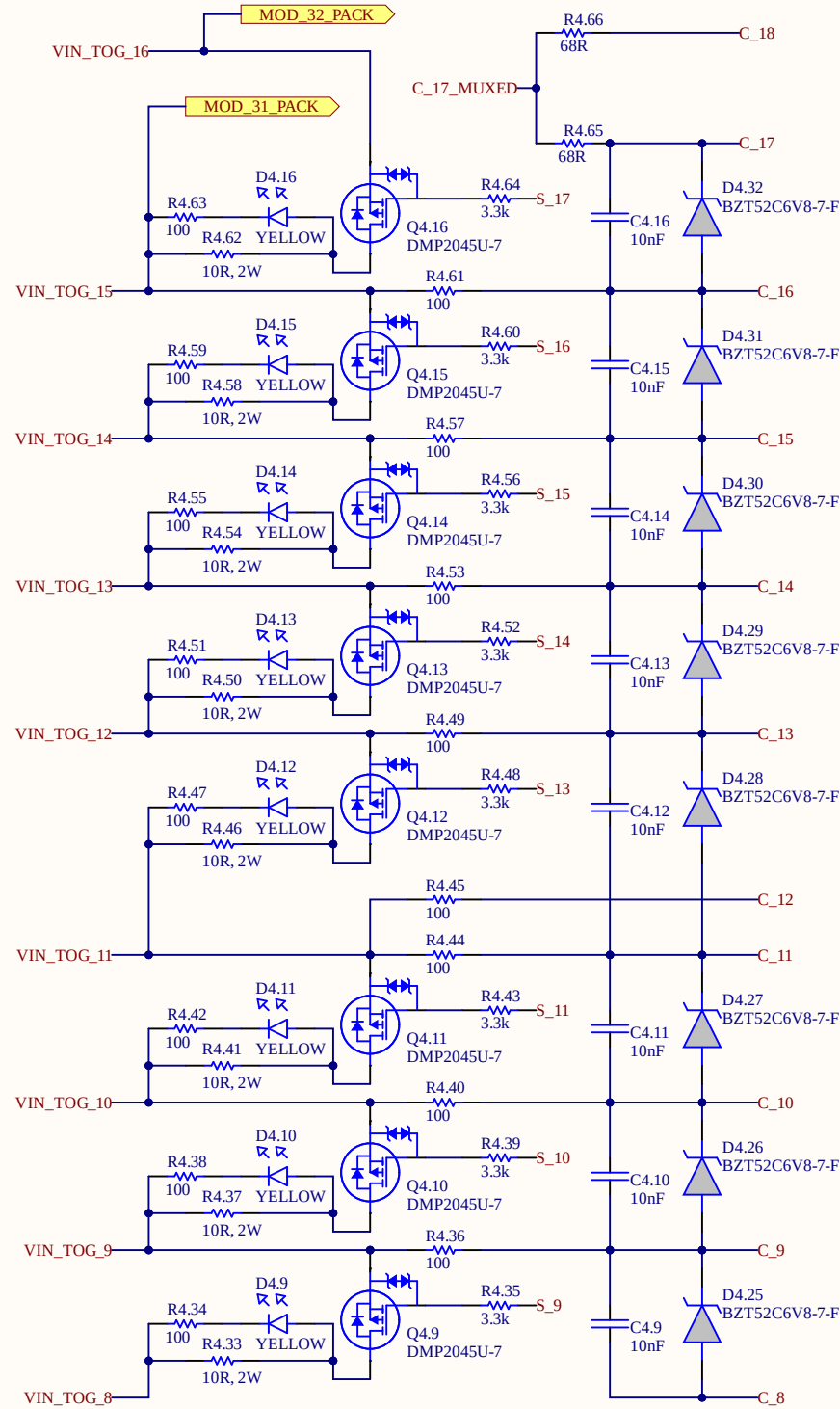
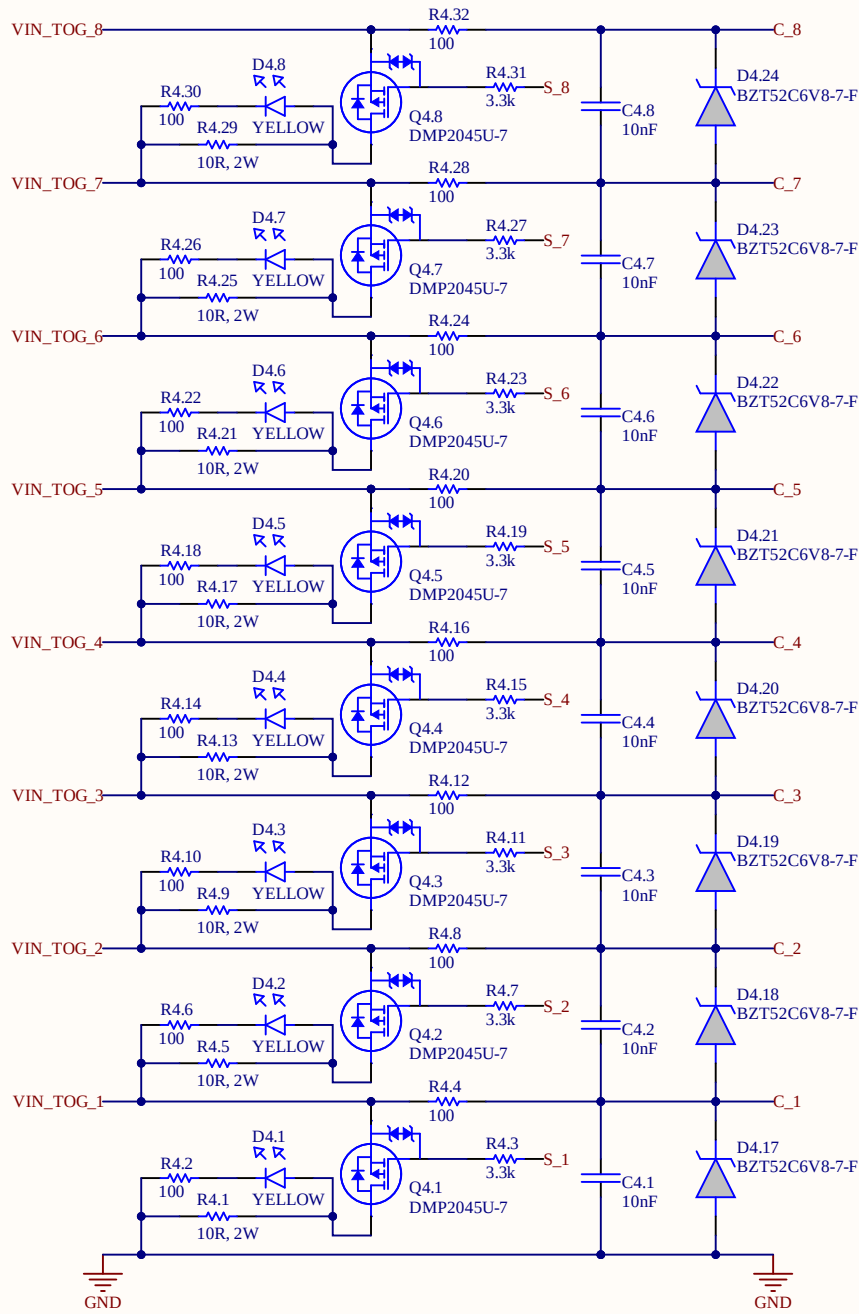
These headers serve a dual purpose as either testpoints or as bypasses for when/if the autoconnection circuitry fails. If used as a bypass, depopulate the FETs and optionally the rest of the circuitry. Then use jumper caps to short the nets together.

We added the gate voltage test point next to these two test points for ease of soldering.

Title		
Size A3	Number	Revision
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File:	Auto_Connection_Circuitry,SchDoc	Drawn By:

4. BALANCING AND FILTERING

^ We have these set to use the DMP2045U-7's for the sake of having ESD protection, however it might be beneficial to use the PJA's simila to the autoconnection circuitry. Both of these (and many similar PFETS) use the same SOT-23 package.



^ C_18 is connected via a SPDT switch to either the last module of the slaveboard (within the pack) or to an external module only used for scrutineering.

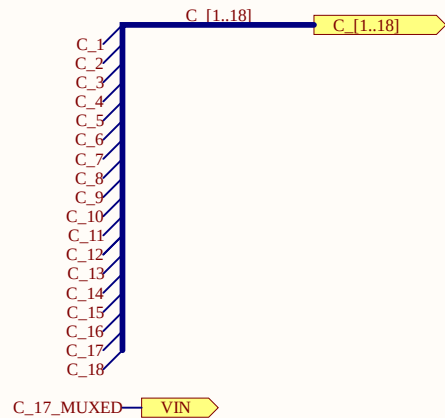
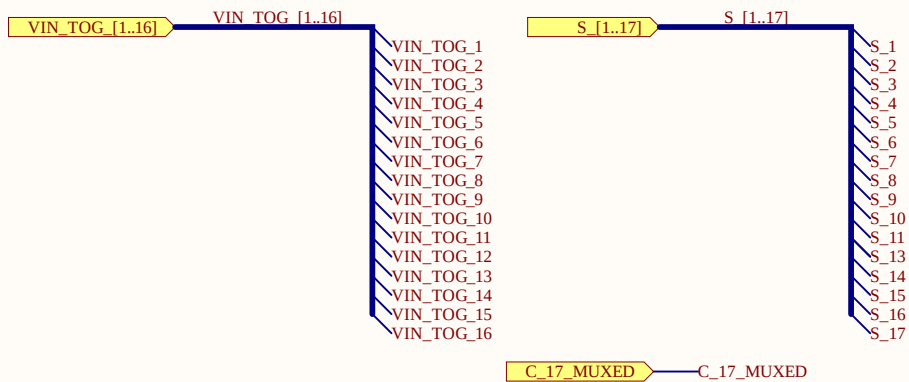
^ The resistance of R8.2 will depend on the exact off resistance of the scrutineering solid state relay. Change based on measurements.

^ C_12 and C_11 are both connected to the 11th module's positive terminal. This is because the datasheet recommends skipping C_12 if using the ADBMS1818 to measure 16 cells. This is for speeding up measurement synchronization as there are only 3 ADCs. See Pg. 86 of the datasheet.

^ Using 6.8V Zener diodes between each input is recommended on Pg. 73 of the ADBMS1818 datasheet Rev. B.

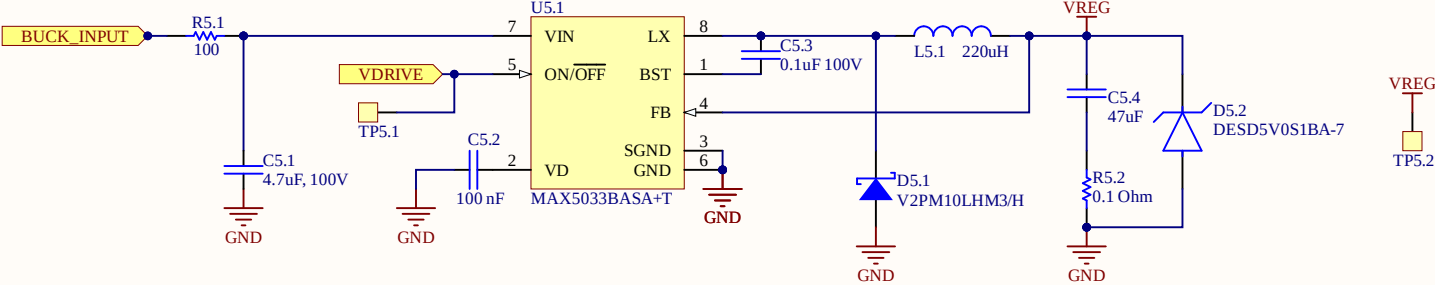
^ The balancing inputs S_1 to S_17 will be pulled down to the previous module voltage when balancing is required, this will close the PFET to being discharge.

Balancing Calculations



Title		
Size	Number	Revision
A3		
Date:	3/22/2026	Sheet of
File:	Balancing_and_Filtering.SchDoc	Drawn By:

5. POWER CONVERTER



The application notes of the datasheet suggesting using a 47uF capacitor, however this requires using an aluminum capacitor which we didn't have space for. 4.7uF should still be sufficient however, according to the component formulas used within the

220uH inductor is used as recommended in the datasheet.

Alternative of 1mH inductor is possible, with a similar footprint.

Component Calculations

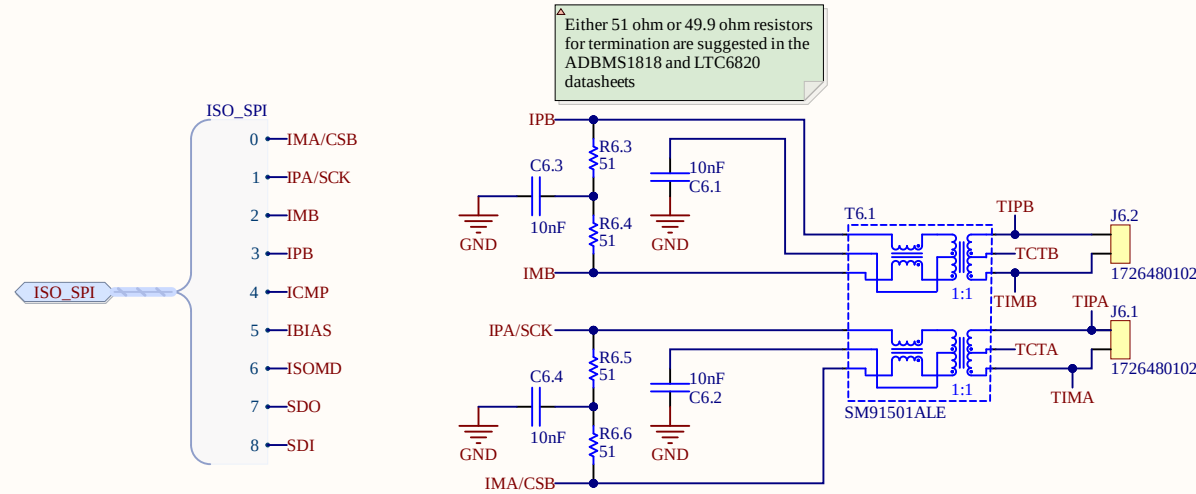
0.1 ohm resistor in series with the output capacitors is used to improve stability.

5.5V TVS diode used to prevent ESD spikes caused by the internals of the buck converter. The maximum input given on the datasheet of the ADBMS for VREG is 5.5V.

0.1 ohm Stability update

Title		
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6. ISOSPI



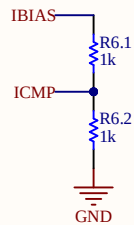
A
 TIMA/TIPA (TP6.4) is the primary SPI input (slave) and will connect to either to the
 masterboard - for first slaveboard
 first slaveboard - for second slaveboard
 TIMB/TIPB (TP6.4) is the secondary SPI input (master) and will connect to either to the
 second slaveboard - for first slaveboard
 N/A - for second slaveboard

- From Pg. 86 of the ADBMS1818 datasheet
- 1. The ADBMS1818 should be 1-2cm away from the transformer
- 2. There should be no ground plane under the transformer or connector
- 3. ISOSPI lines should be kept as direct as possible and not cross over any other signal/analog lines

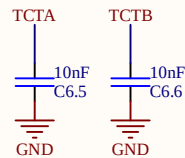
1k resistors are used to have a threshold voltage of 1V and an IsoSPI current of 20mA.

See calculations here:

IsoSPI bias resistor calculations



Extra center-tap bypass capacitors for further reducing common mode noise if needed

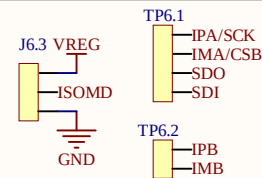


▲ Pull up on SDO line for using 4-wire SPI (as suggested in datasheet)

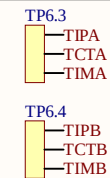


ISOMD needs to be shorted to VREG normally. During testing we can short ISOMD to GND to activate 4-wire SPI mode which we can use for testing.

During this testing, we will use **TP6.4** to connect the SPI of the masterboard to the slaveboard via 4 female-female jumper wires.



These ISOSPI test points can be used to probe the isoSPI lines with an oscilloscope for testing.



△ DNP TP6.1, TP6.2, TP6.3, TP6.4, unless necessary. This is because the testpoints form antennas for these high speed signals (not good).

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Size A4	Number	Revision
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File:	IsoSPI.SchDoc	Drawn By:

7. TEMPERATURE MUX

▲ Pull down resistors for T_SEL to put the multiplexer into a valid state if nothing driving the line

T_SENSE_SCRUT

T_SENSE_[1..15]

- T_SENSE_1
- T_SENSE_2
- T_SENSE_3
- T_SENSE_4
- T_SENSE_5
- T_SENSE_6
- T_SENSE_7
- T_SENSE_8
- T_SENSE_9
- T_SENSE_10
- T_SENSE_11
- T_SENSE_12
- T_SENSE_13
- T_SENSE_14
- T_SENSE_15

▲ T_READ inputs are multiplexed outputs from the 16 temperature sensing channels. We multiplex between these using the T_SEL outputs from the GPIO's of the ADBMS

Temperature_Sense

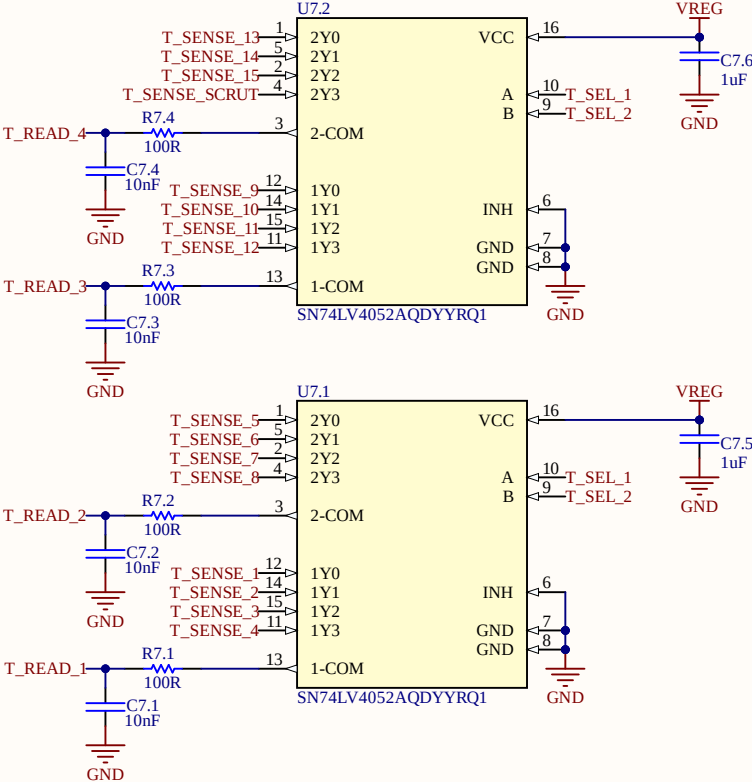
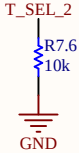
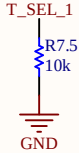
- 1 → T_READ_1
- 2 → T_READ_2
- 3 → T_READ_3
- 4 → T_READ_4

T_SENSE

T_Select

- 1 → T_SEL_1
- 2 → T_SEL_2

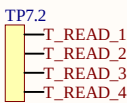
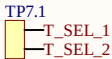
T_SELECT



▲ The Multiplexer datasheet suggests to keep the signal lines as short and straight as possible, for our purposes this isn't that necessary as the lines are not that sensitive.

However, we should keep the VREG capacitors as close as possible to the MUX as possible

▲ We only include test points for selecting and reading the temperature, as the testing PCB will allow us to probe the individual T_SENSE pins

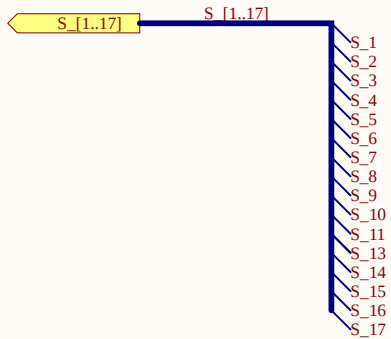
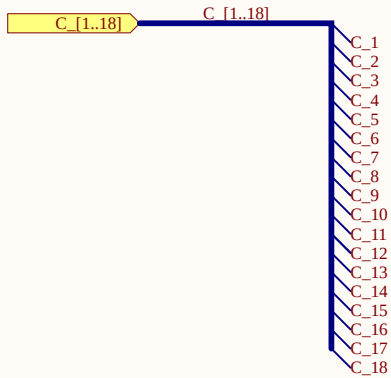


▲ Capacitor values for VREG and T_READ were somewhat arbitrarily chosen as 1uF and 10nF to ensure that these voltage nets stay somewhat constant respectively.

VREG is the power suply line and thusse requires higher capacitance of 1uF. The analog RC filter values were set based on the ADC characteristics in the datasheet.

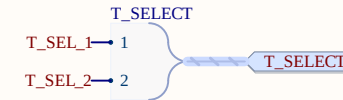
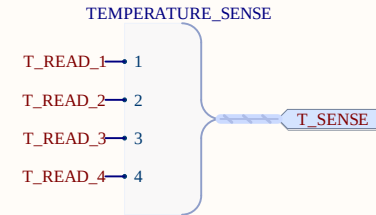
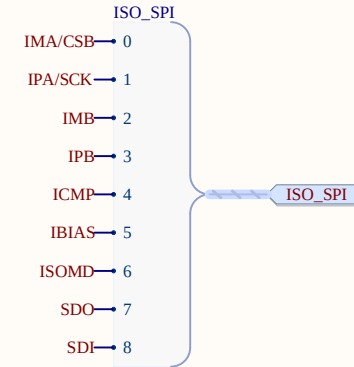
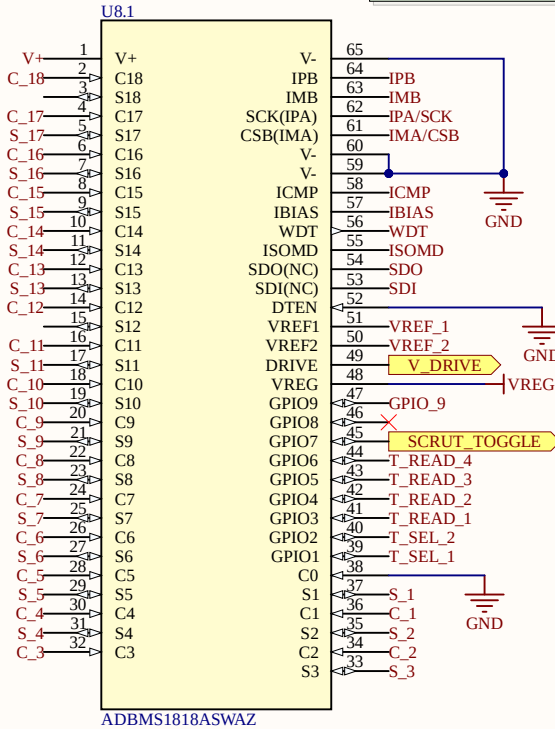
Title		
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File:	Temperature_Circuitry.SchDoc	Drawn By:

8. ADBMS1818



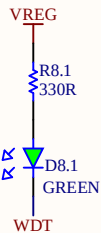
ADBMS1818

Note: We can revert to the LTC6813 as well, since it has the same footprint.



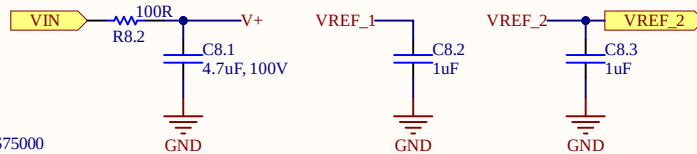
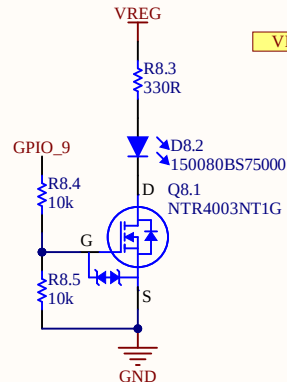
Capacitor values for VREF1, VREF2, and VIN taken from Pg 89 of the ADBMS1818 datasheet

Testpoints, and Watchdog LED.



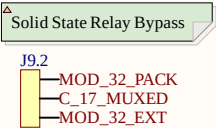
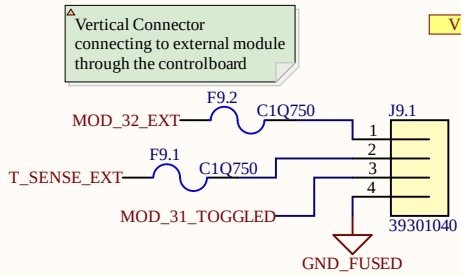
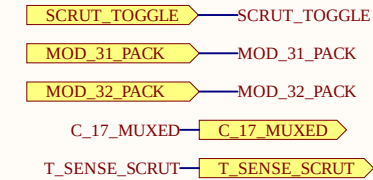
DTEN is pulled to GND to disable the discharge timer (we don't use it).

WDT is pulled to GND when the ADBMS is active, and then when the ADBMS goes to sleep (Watchdog timer runs out) the WDT will be pulled to floating. (~2 seconds)



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File:	ADBMS1818.SchDoc	Drawn By:

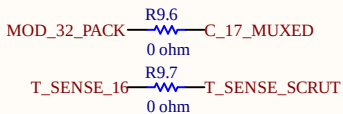
9. SCRUTINEERING



If the solid state relays circuitry fails, leave both solid state relays unpopulated and use these jumpers to short C_17 to either the scrutineering external module or internal module

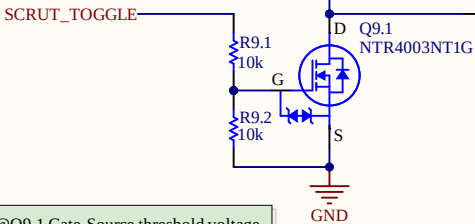
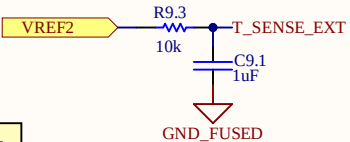


0 ohm resistors to bypass scrutineering circuitry for slaveboard #1

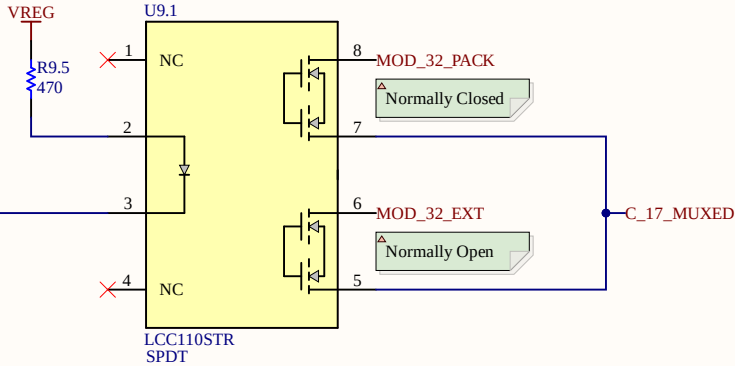


Note: Only populate Solid @R9.6 and @R9.7 for slaveboard #1 (mod 1-16) DO NOT POPULATE FOR SLAVEBOARD #2 (mod 17-32)

Temperature low pass filter



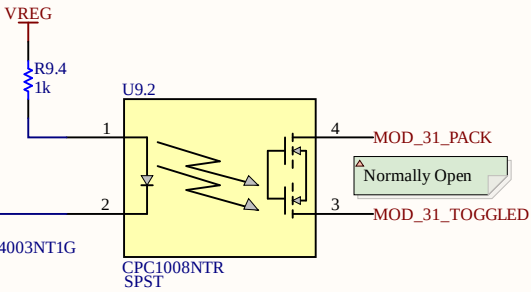
Note: Only populate Solid State Relays @U9.1 and @U9.2 for slaveboard #2 (mod 17-32), slaveboard #1 (mod 1-16) doesn't need to have scrutineering components.



@U9.1 is a SPDT relay used for connecting C18 to either the external scrutineering module or to the internal 32nd module.

If SCRUT_TOGGLE is high (5V) the NFET will close, allowing current through the solid state relay. This will short @MOD_32_EXT to C_17_MUXED.

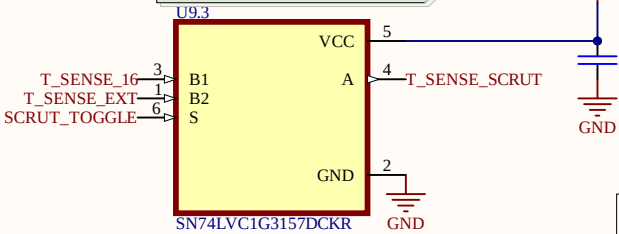
If SCRUT_TOGGLE is low, MOD_32_PACK will be shorted to C_17_MUXED.



@U9.2 is a SPST relay used for only sending the 32nd module voltage reference to the controlboard if scrutineering mode is active. This is because we only want to supply this high voltage to the controlboard when its being used.

If SCRUT_TOGGLE is high (5V) the NFET will close, allowing current through the solid state relay. This will short MOD_32_PACK to MOD_32_TOGGLED

Multiplexer for the 16th Temperature sense channel.



@U9.3 is a 2 -> 1 multiplexer used for selecting the temperature sensing channel for the 16th module (either the in pack module or the external scrutineering modules temperature sense).

If SCRUT_TOGGLE is high (5V) then the multiplexer will use B2 which is the external module, if SCRUT_TOGGLE is low (0V) then the multiplexer will use B1 (the internal module).

Title		
Size	Number	Revision
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Date:	3/22/2026	Sheet of
File:	Scrutineering_Circuitry.SchDoc	Drawn By: